

Figure 5: VM groups



Figure 7: Guest addition installation

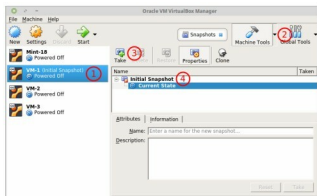


Figure 6: Snapshot VM

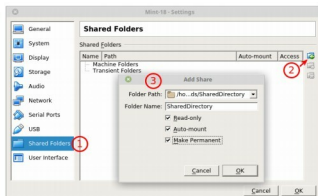


Figure 8: Shared folder

Not Attached, NAT, bridged adapters, internal networks and host-only adapters.

Perform the steps below to view/manipulate the current network settings:

- 1) Select the VM from the VirtualBox Manager.
- 2) Click the 'Settings' button on the toolbar.
- 3) Select the 'Network' option from the left pane.
- 4) Select the adapter. The current networking mode will be displayed under the 'Attached to' drop-down box.
- 5) To change the mode, select the required network mode from the drop-down box and click the 'OK' button.

Figure 9 illustrates the above steps.

VirtualBox network modes

Let us discuss each network mode briefly.

- 1) **Not Attached:** In this mode, VirtualBox reports to the guest that the network card is installed but it is not connected. As a result of this, networking is not possible in this mode. If you want to compare this scenario with a physical machine, then it is similar to the Ethernet card being present but the cable not being connected to it.
- 2) **NAT:** This stands for Network Address Translation and it is the default mode. If you want to access external

networks from the guest, then this will serve your purpose. It is similar to a physical system connected to an external network via the router.

- 3) **Bridged adapter:** In this mode, VirtualBox connects to one of your installed network cards and exchanges network packets directly, circumventing the host operating system's network stack.
- 4) **Internal:** In this mode, communication is allowed between a selected group of VMs only. Communication with the host is not possible.
- 5) **Host only:** In this mode, communication is allowed between a selected group of VMs and the host. A physical Ethernet card is not required; instead, a virtual network interface (similar to a loopback interface) is created on the host.

An introduction to VBoxManage

VBoxManage is the command line interface (CLI) of VirtualBox. You can manage VirtualBox from your host via these commands. It supports all the features that are supported by the GUI. It gets installed by default when the VirtualBox package is installed. Let us look at some of its basic commands.